

Genie API QPM Planner

User Input - Please Modify as Needed
Calculations - Please Do NOT Change
Resulting QPM / Utilisation Metrics - Please Do NOT Change

Usage assumptions (per hour)
Define how your users behave during an average and peak hour. These inputs shape the base demand before modelling the full day.

Average users per active hour

The typical number of users active in a non-peak hour. Used to estimate baseline API/Genie load.

10

Peak users in busiest hour

The maximum number of users active in the busiest hour of the day (your highest concurrency). Determines peak QPM.

50

Average visit duration (minutes)

How long a typical user session lasts. Longer sessions → more concurrent usage.

10

Adoption rate (%)

The percentage of your total population that actually uses the system each hour. Helps scale down large user counts realistically.

50.00%

Number of peaks per weekday (1, 2, or 3 supported)

How many traffic spikes your application has per day (e.g., 1 = midday; 2 = morning + afternoon; 3 = morning + noon + evening). Shapes the daily demand curve.

2

Peak width (hours)

Controls how sharp or wide each peak is. Smaller values = sharper spikes. Larger values = broader, flatter peaks.

3

Interaction behaviour

Control how quickly your users issue queries. Scenarios affect overall QPM load.

Chosen scenario

Selects a predefined interaction style (Fast, Standard, Slow).

Standard

Selected time per query (seconds)

Automatically filled based on the chosen scenario.

Represents how long an average interaction takes → affects QPM.

90

Scenario

A scenario represents how quickly your users interact with Genie/API. Specifically, how many seconds it takes a user to submit a query, read the answer, and submit the next one.

Fast

Standard

Slow

Time per query (s)	Avg QPM	Peak QPM
45	1.11	5.56
90	0.56	2.78
150	0.33	1.67

Weekend behaviour

Weekend usage (% of weekday)

20.00%

SLA / business preference

Target % of queries WITHOUT queuing

Your desired SLA. Used only for guidance → not part of core calculations.

99.99%

QPM reservation

The QPM limit you plan to reserve/purchase.

All queuing, utilisation, and "time above cap" metrics are calculated using this value.

Derived metrics

Calculated values that describe your expected demand profile.

Average concurrent users

The estimated number of users simultaneously active at any given moment during a typical hour. Calculated from: (Average users per hour × Average session duration) / 60

0.83

Peak concurrent users

The maximum number of users simultaneously active during the busiest hour of the day. This determines your theoretical peak demand before scaling by interaction behaviour.

4.17

Avg QPM (selected scenario)

The average number of queries per minute generated during normal load based on your selected interaction style.

0.56

Peak QPM (selected scenario)

The expected number of queries per minute during the busiest period of the day, reflecting both peak concurrency and interaction speed.

2.78

For chosen QPM reservation

Shows how your selected reservation performs under your traffic model.

% queries queued

The percentage of total daily queries that exceed your reserved QPM capacity and must wait (i.e., should be put in a queue).

0.00%

% time above cap

The percentage of the day during which demand exceeds your reserved QPM.

0.00%

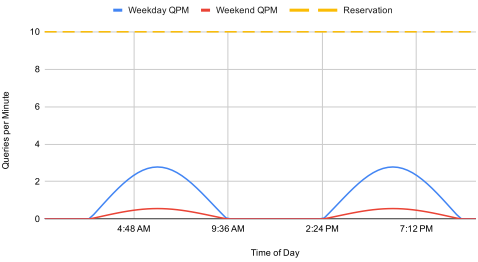
Average utilisation

How much of your reserved QPM is actually used on average across the entire day (weekday + weekend).

7.73%

Daily QPM Load Profile

How the distribution is model on a daily basis (weekday, weekend, reservation line)



Queueing vs QPM Reservation

Shows how different reservation levels affect queueing and time-over-cap. Use this to find the optimal

